

<!DOCTYPE html>

ETSI GS QKD 004 push mode specification, QKD server

```
<div id="redoc"><div class="sc-dvwKkn kQuWHA redoc-wrap"><div class="sc-eVQf1l coPBSC menu-content" sty
  m -673.67664,1221.6502 -231.2455,-231.24803 55.6165,
  -55.627 c 30.5891,-30.59485 56.1806,-55.627 56.8701,-55.627 0.6894,
  0 79.8637,78.60862 175.9427,174.68583 1 174.6892,174.6858 174.6892,
  -174.6858 c 96.079,-96.07721 175.253196,-174.68583 175.942696,
  -174.68583 0.6895,0 26.281,25.03215 56.8701,
  55.627 1 55.6165,55.627 -231.245496,231.24803 c -127.185,127.1864
  -231.5279,231.248 -231.873,231.248 -0.3451,0 -104.688,
  -104.0616 -231.873,-231.248 z
  " fill="currentColor"></path></g></svg><svg class="" style="transform:translate(2px, 4px);transitio
  m -673.67664,1221.6502 -231.2455,-231.24803 55.6165,
  -55.627 c 30.5891,-30.59485 56.1806,-55.627 56.8701,-55.627 0.6894,
  0 79.8637,78.60862 175.9427,174.68583 1 174.6892,174.6858 174.6892,
  -174.6858 c 96.079,-96.07721 175.253196,-174.68583 175.942696,
  -174.68583 0.6895,0 26.281,25.03215 56.8701,
  55.627 1 55.6165,55.627 -231.245496,231.24803 c -127.185,127.1864
  -231.5279,231.248 -231.873,231.248 -0.3451,0 -104.688,
  -104.0616 -231.873,-231.248 z
  " fill="currentColor"></path></g></svg></div></div><div class="sc-jtcaXa kl0bZX api-content"><div c
" class="sc-iIPly sc-cCsOjm bAuwZn grHHWS">
```

Server description of the QKD endpoint processing ETSI GS QKD 004 push mode requests

etsi004

open key stream

The open-connect message connects a KMS to the QKD Device. Conforming to the specification, the QKD device will assign the `key_stream_id` and can suggest alternative QoS due to its capabilities.

Request Body schema: application/json

required

source

required

string <uri>

the source URI of the request.

destination

required

string <uri>

the destination URI of the request.

key_stream_id

string <uuid>

key stream ID. If not specified it is up to the QKD device to define the UUID

qos

required

object (qos)

requested QoS

Responses

201

Key stream successfully created

400

Client sent a malformed input

500

The server experienced an internal error

post/open-connect

https://qkd_server/api/v1/kms/etsi004/open-connect

Request samples

Payload

Content type

application/json

Copy

Expand all

Collapse all

```
{
  "source": "http://kms-1.net",
  "destination": "http://kms-2.net",
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20",
  "qos": {
    "key_chunk_size": 32,
    "max_bps": 128,
    "min_bps": 64,
    "jitter": 1,
    "priority": 3,
    "timeout": 50,
    "ttl": 600,
    "metadata_mimetype": "application/json"
  }
}
```

Response samples

201

Content type

application/json

Copy

Expand all

Collapse all

```
{
  "source": "http://kms-1.net",
  "destination": "http://kms-2.net",
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20",
  "qos": {
    "key_chunk_size": 32,
```

```

“max_bps”: 128,
“min_bps”: 64,
“jitter”: 1,
“priority”: 3,
“timeout”: 50,
“ttl”: 600,
“metadata_mimetype”: ”application/json”
},
“status”: 0
}

```

start pushed key retrieval process

The get-key message is the starting point after which the QKD device can start pushing keys.

Request Body schema: application/json

required

key_stream_id

required

string <uuid>

corresponding to the one given in open_connect

index

integer <uint32>

increasing index of the key chunk in the key stream. Wrapping at UINT_MAX.

metadata

object (metadata)

Responses

205

Successful registration

400

Client sent a malformed input

500

The server experienced an internal error

get/get-key

https://qkd_server/api/v1/kms/etsi004/get-key

Request samples

Payload

Content type

application/json

Copy

Expand all

Collapse all

```

{
“key_stream_id”: ”4eb81a5c-031f-4d1f-881d-309bee44fc20”,

```

```
“index”: 0,  
“metadata”: {  
“size”: 0,  
“buffer”: ””  
}  
}
```

Response samples

205

Content type

application/json

Copy

Expand all

Collapse all

```
{  
“status”: 0,  
“metadata”: {  
“size”: 0,  
“buffer”: ””  
}  
}
```

close the key stream

The close message disconnects the KMS from the QKD device.

Request Body schema: application/json

required

key_stream_id

required

string <uuid>

corresponding to the one given in open_connect

Responses

204

Key stream successfully closed

400

Client sent a malformed input

500

The server experienced an internal error

put/close

https://qkd_server/api/v1/kms/etsi004/close

Request samples

Payload

Content type

application/json

Copy

```
{  
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20"  
}
```

Response samples

204

Content type

application/json

Copy

```
{  
  "status": 0  
}  
  
<script>  
  const __redoc_state = {"menu":{"activeItemId":-1},"spec":{"data":{"openapi":"3.1.2","info":{"title":  
  
    var container = document.getElementById('redoc');  
    Redoc.hydrate(__redoc_state, container);  
  
  </script>
```